22 A.P.J. Abdul Kalam

Abul Pakir Jainulabbdin Abdul Kalam, 'the missile Man of India' and President of India was born in a humble family in Rameshwaram, Tamilnadu. The teetotaler and vegetarian and bachelor Dr Kalam has tremendous faith in the youth of India. He says, 'We have many young people of high calibre. All that they need is a dream to chase and fulfill. We should leave mediocrity behind and always aim for the best.'

Besides being a scientist of high repute Dr Kalam is a poet and an avid lover of music. Although he has very few leisure moments in his busy schedule,he spends his leisure time playing the veena. Technology is a buzzword for him. But then he is a scientist who thinks dreams, lives and breathes science. What makes Dr Kalam great is his austere and disarming simplicity notwithstanding his great achievements and honours in the field of science. It was Dr Kalam who as the Director of Defence Research and Development Laboratory lined up Prithvi, Agni, Trishul, Akash and Nag. He was conferred numerous awards including Padma Bhushan and Bharat Ratna. All these laurels were earned by his unremitting toil and unflinching devotion and hard work. He enthused scientists and infused a fresh whiff of informality into the laboratory which had a military aura around it. He believed in the team spirit. His dream materialised, thanks to the gruelling schedule of working for eighteen hours a day.

Apart from developing missiles and satellites, Dr Kalam brought smiles on the faces of handicapped children by designing ultra light weight foot support for polio victims. The polio-affected children will be grateful to him for the light weight calipers.

A spiritual and humble son of the soil is a scintillating example of success through self-restraint. He stoutly believes, 'our youth must dream, dream, dream; convert those dreams into thoughts; and them transform these thoughts into action. We must think big.' This committed man of India is to make the nation self-reliant in military technology.

Work Brings Solace

On my return from France, after successfully testing the SLV–3 apogee motor, Dr Brahm Prakash informed me one day that Wernher von Braun would be visiting Thumba. I was to pick him up in Madras.

Everybody working in rocketry knows of von Braun, who made the lethal *V–2 missiles* that devastated London in World War II. In the final stages of the war, von Braun was captured by the Allied forces. As a tribute to his genius, von Braun was given a top position in the rocketry programme at NASA. Working for the US Army, von Braun produced the landmark *Jupiter* missile, which was the first IRBM with a 3,000–km range.

The *V–2* missile was by far the greatest single achievement in the history of rockets and missiles. It was the culmination of the efforts of von Braun and his team in the VFR (Society for Space Flight, in Germany) in the 1920s. What had begun as a civilian effort soon became an official army one, and von Braun became the technical director of the German Missile Laboratory at Kummersdorf. The first *V–2* missile was tested unsuccessfully in June 1942. It toppled over on to its side and exploded. But on 16 August 1942, it became the first missile to exceed the speed of sound.

It filled me with awe that I would be travelling with this man—a scientist, designer, production engineer, administrator and technology manager, all rolled into one.

The *Avro* aircraft I brought him in took around ninety minutes from Madras to Trivandrum. Through the flight, von Braun asked me about our work and listened as if he was just another student of rocketry. I never expected the father of modern rocketry to be so humble, receptive and encouraging. He made me feel comfortable right through the flight.

How did he feel in the US? I asked this of von Braun, who had become a cult figure in America after creating the *Saturn* rocket in the *Apollo* mission (which put the first man on the moon).

"America is a country of great possibilities, but they look upon everything un–American with suspicion and contempt. They suffer from a deep–rooted NIH—Not Invented Here—complex, and look down on alien technologies. If you want to do anything in rocketry, do it yourself," von Braun advised me.

He continued, "SLV–3 is a genuine Indian design and you may be having your own troubles. But you should always remember that one doesn't just build on successes, but also on failures. Mere hard work cannot fetch you honour. Building a rock wall is backbreaking work. There are some people who build rock walls all their lives. And when they die, they leave behind miles of walls, mute testimony to how hard they had worked.

"But there are others who, while placing one rock on top of another, have a vision. It may be to create a terrace with roses climbing over the rock walls and chairs set out for lazy summer days. Or the rock wall may enclose an apple orchard or mark a boundary. What matters is that when they finish, they have more than a wall. It is this goal that makes the difference.

"Did not make rocketry your profession or your livelihood—make it your religion, your mission."

Did I see something of Prof. Vikram Sarabhai in Wernher von Braun? It made me happy to think so.

With three deaths in the family in as many years, I needed total commitment to my work in order to keep performing. I wanted to throw all my being into creating the SLV. During this period, it was as though I had pushed a 'hold' button on my life—no badminton in the evenings, no more weekends or holidays,, no family, no relations, not even any friends outside the SLV circle.

To succeed in your mission, you must have single–minded devotion to your goal. Individuals like my self are often called 'workaholics'. I question this term because that implies an illness. If I work towards that which I desire more than anything else in the world, and which also makes me happy, how can it be considered an aberration?

The desire to work at optimum capacity leaves hardly any room for anything else. I have had people with me who would scoff at the 40–hours–a–week job they were paid for. I have also known others who worked 60, 80 and even 100 hours a week because they found their work exciting and rewarding. Total commitment is the common denominator among all successful men and women.

Once you have done this—charged yourself, as it were, with your commitment to your work—you also need good health and boundless energy. Climbing to the top demands strength—whether to the top of Mount Everest or to the top of your career.

'Flow' is an overwhelming and joyous experience while working—it is a sensation we experience when we act with total involvement. During flow, action follows action according to an internal logic; there seems to be no need of conscious intervention on the part of the worker. There is no hurry and there are no distracting demands on one's attention. The distinction between self arid the activity disappears.

All of us working on SLV were experiencing flow. Although we were working very hard, we were very relaxed, energetic and fresh. How did it happen? Who had created this flow? Perhaps it was the fact that the difficult targets we had set actually seemed achievable.

When the SLV–3 hardware started emerging, our ability to concentrate increased markedly. I felt a tremendous surge of confidence—I was in complete control over myself and over the SLV–3 project.

The first requirement to get into flow is to work as hard as you can at something that presents a challenge. It may not be an overwhelming challenge, but one that stretches you a little; something that makes you realise that you are performing a task better today than you did yesterday, or the last time you tried to do it.

Another pre–requisite for being in flow is the availability of a significant span of uninterrupted time. In my experience, it is difficult to switch into the flow state in less than half an hour. And it is almost impossible if you are disturbed constantly.

I have experienced this state many times, almost every day of the SLV–3 mission. There have been days in the laboratory when I have looked up to find the laboratory empty and realised that it was way past my work hours. On other days, my–team members and I have been so caught up in our work that the lunch hour slipped by without our even being conscious that we were hungry.

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About the Essay

The former President of India who is popularity known as 'the missile man' describes in 'Work brings Solace' two things. The first is Werner von Braun, the great scientist who made the lethal V-2 missile that devastated London in World War II and who was given a top position in the rocketry programme at NASA. Dr Kalam was very much impressed by his humble, receptive and encouraging quality. Dr Kalam has also quoted his remarks about America.

The second thing that is stressed is the total surrender to work gives immense pleasure and satisfaction. Generally professional personalities perform 40-hours a week job but when the work is exciting, rewarding and with full of passions even 100 hours a week job does not become strenuous but keep one relaxed, energetic and fresh. Dr Kalam has described his personal experiences how he kept himself away from badminton, holidays, family relations, friends out side the SLV circle in order to achieve the desired goal.

Glossary

apogee: the most successful, popular or powerful point
rocketry: to rise extremely, quickly or make extremely quick progress towards success
lethal: able to cause or causing death
culmination: the end of something, especially if it comes after a long time
aberration: a temporary change from the typical or usual way of behaving
denominator: the number below the line in a fraction